

Sugar cane shoot borer	<i>Chilo infuscatellus</i>	Pyralidae	Lepidoptera
Gurdaspur borer	<i>Acigona steinella</i>	Pyralidae	Lepidoptera
Sugar cane internode borer	<i>Chilo sacchariphagus indicus</i>	Pyralidae	Lepidoptera
Sugar cane scale insect	<i>Melonapsis glomerata</i>		

Insect -Pests of Pulses

Common name	Scientific name	Family
Gram pod borer	<i>Helicoverpa armigera</i>	Noctuidae
Gram cut worm	<i>Agrotis ipsilon</i>	Noctuidae
Red gram pod fly	<i>Melanagromyza obtusa</i>	Agromyzidae
Gram pod borer	<i>Helicoverpa armigera</i>	Noctuidae
Gram cut worm	<i>Agrotis ipsilon</i>	Noctuidae

1. Gram pod borer: *Helicoverpa armigera*

- **Family:** Noctuidae
- **Order:** Lepidoptera
- It is polyphagous & cosmopolitan in nature.
- Caterpillar feed foliage, when young and on seeds in the later stages.
- Larvae pupate in soil.
- ❖ **Management:**
 - Timely sowing (up to mid-October).
 - Grow early maturity varieties.
 - Mixed/intercropping with barley, wheat, mustard.
 - Application of NPV @ 250-500 LE/ha. (RPSC, A0-09)
 - **Egg parasitoid** *Trichogramma chilonis*.

- **Larval parasitoid** of *Helicoverpa* is *Cosmopletis chloridae*.
- Use of light traps and Pheromone traps.
- Application of Monocrotophos @ 1.25 liter/ha at evening time.

2. Gram cut worm: *Agrotis ipsilon*

- **Family:** Noctuidae
- **Order:** Lepidoptera
- Polyphagous pest but cause much damage to gram crop.
- Cannibalism habit present in larvae.
- The young larvae feed on the epidermis of leaves.
- During daytime larvae live in cracks and holes in ground and come out at night and fall the plants by cutting their stem either below the surface or above the ground.

❖ **Management:**

- Hand picking & destruction of larvae.
- Application of Flubendiamide (Fume) @ 0.5 % sprays at evening time.
- Application of Malathion 5 % dust @ 25 kg/ha at evening time

3. Red gram pod fly: *Melanagromyza obtuse*

- **Family:** Agromyzidae
- **Order:** Diptera.
- The maggots eat away only a part of the seed.
- The partially damaged seeds become subjected to bacterial and fungal infection.
- Grains are rendered, unfit for human consumption.
- Maggots pupate in pods.

❖ **Management:** Application of Dimethoate @ 1.0 liter in 250-liter water/ha.

4. Other pests of Pulses:

Common name	Scientific name	Family	Order	Remarks
Tur pod bug	<i>Clavigralla gibbosa</i>	Coreidae	Hemiptera	Both nymph and adult suck the cell sap on leaves and pods of Arhar, cow pea and gram
Plume moth	<i>Exelastis atomosa</i>	Pterophoridae	Lepidoptera	It is specific pest of Arhar. Caterpillar scrapes of pods and make hole on pods.
Girdle beetle	<i>Oberia bravis</i>	Cerambycidae	Coleoptera	It is specific pest of soybean. The ovipositing female beetle girdles the stem twice. It makes three punctures just above the lower ring before inserting a single egg through the largest hole into pith. Management: spray methyl parathion 50 EC 1.25 liter/ha.
Bean fly	<i>Ophiomyia phaseoli</i>	Agromyzidae	Diptera	Feed on Phaseolus, cow pea, soybean, Arhar
Lentil pod borer	<i>Etiella zinckenella</i>	Phycitidae	Lepidoptera	

Diseases and Enemies of Honeybees

1. Predatory wasps:

<i>Vespa auraria</i>	Nests on treetops/buildings
<i>Vespa magnifica</i>	Under-ground nest
<i>Vespa cincta</i>	Underground nest
<i>Vespa basalis</i>	Nest on treetop/buildings

- The wasps catch the bees at hive entrance and kill them.
- Most serious damage in hills is caused by *V. magnifica* which cuts down bees in large number.
- The weak colonies may even perish due to its attack.
- Burn the nests during nighttime.
- Kill the wasps in the apiary by flapping

❖ **Wax moth (*Galleria mellonella*)**